JEYARAJA KARTHIKEYAN A S

Veppadai, Namakkal Dt, Tamil Nadu, India | 9445935350 | jeyarajakarthikeyan@gmail.com

OBJECTIVE

Aspiring Mechatronics Engineer seeking an opportunity to improve my design skills, gain industrial experience, and grow as an engineer. I want to work on real projects, learn from experienced professionals, and build a stable and successful career.

EDUCATION

Sona College of Technology — B.E. Mechatronics Engineering *2023–2027*

CGPA: 7.52

KSR Akshara Academy — Senior Secondary (Grade 12) *2023*

Score: 76%

Sri Vani International CBSE School — SSLC (Grade 10) *2021*

Score: 81.6%

EXPERIENCE

Engine Factory, Avadi | Production & Quality Control *Dec 2025*

- Developed CAD design for pressure testing fixture for MBT engine crankshafts
- Acquired knowledge of defense-grade manufacturing standards and quality inspection methods

Uno Minda Limited (Seating Division) | Production & Manufacturing *Jul 2025*

- Gained exposure to end-to-end automotive seating production cycle
- Documented industrial sensors and control systems applications in high-output manufacturing

PROJECTS

Adaptive Smart Ventilation System – IoT-Based Ventilation Control

- ESP32-based system using environmental and occupancy sensors with adaptive control logic for efficient, automated ventilation
- Role: Team leader - Project Planning, Team Coordination, Task Delegation & Problem Solving

MotorSense – AI-Powered Predictive Maintenance

- IoT and AI-driven platform using ESP32, MPU9250, ACS712, DS18B20 sensors with Isolation Forest ML algorithm for real-time machine health monitoring and anomaly detection
- Role: Research, Report Preparation, Presentation & Project Demonstration

Self-Healing IoT PCB

- 3-Plane autonomous fault management system with ESP32, eFuses, and Blynk cloud telemetry for runtime degradation detection
- Role: Research, Report Preparation, Presentation & Project Demonstration

IoT-Based Posture Monitoring System

- Smart ergonomic solution using ESP32, FSR sensors, automated air-pillow actuation, and haptic feedback for real-time posture correction
- Role: Team leader - Project Planning, Team Coordination, Task Delegation & Problem Solving

SKILLS

CAD/Design: SolidWorks, 3DEXPERIENCE (3DX)

Programming: C++, Python, SQL

INTERESTS

CAD and Mathematics

ACHIEVEMENTS

- 1st Place — Innovatronix Expo 2026: MotorSense – AI-Powered Predictive Maintenance, Dept. of Mechatronics & ATOM Student Association, SONA College of Technology (16 Apr 2026)
- 1st Place — Digi X Thon 2026: Smart Electronics IoT Project – Self-Healing IoT PCB, Dept. of ECE, Dr. M.G.R. Educational & Research Institute (30 Mar 2026)
- 2nd Place — Pratiyogita Volume 2 – Code Challenge, IEEE & Kongu Engineering College (23–24 Jan 2026)
- 3rd Place — Robotroniks 2024 - Group Discussion, Dept. of Mechatronics, SONA College of Technology (28 Sep 2024)

CERTIFICATIONS

- Design Thinking – NPTEL
- Automation in Manufacturing – NPTEL
- Acquiring Data – IT-ITeS Sector Skills Council (NASSCOM), Mar 2026
- Digital Edge 101 – IT-ITeS Sector Skills Council (NASSCOM), Jan 2026

ACTIVITIES

- TN-IMPACT 2026 — Project Presentation (14 Mar 2026)
- Pratiyogita Volume 2 Techfest — Project Presentation & Sensor Challenge, IEEE & Kongu Engineering College (23 Jan 2026)
- Sona Techathon'25 — Project Presentation, Dept. of Training & Placement / Sona Innovation Centre (8 May 2025)
- KRIZEN'24 — Project Presentation, Sri Krishna College of Engineering & Technology (18 Oct 2024)
- Spring Fest 2K24 — Project Expo & Technical Quiz, K.S.R. College of Engineering (27 Sep 2024)
- Organizer — English Club, Sona Modern Theatre

KEY STRENGTHS

Quick Learning • Adaptable • Problem Solving • Continuous Learner • Analytical Thinker